

Servas Adolph Tarimo

Student and Researcher
Department of Future
Convergence Technology.
Soonchunhyang University
servasadolph@sch.ac.kr
servasadolph@gmail.com
servasadolph09@gmail.com
(Last Updated: Sept. 09, 2024)

PERSONAL INFORMATION

- **Birth:** September 09, in Tanzania.
- **Nationality:** Tanzanian
- **Language:** Swahili (Native), English (Intermediate)
- **Status:** Single

PROFESSIONAL SUMMARY

Servas Adolph is a PhD student in the Department of Future Convergence Technology at **Soonchunhyang University**, South Korea. With a Master's degree in Big Data Engineering and a Bachelor's in Computer Engineering and Information Technology, his research focuses on **multimodal machine learning for healthcare, health data analytics, and artificial intelligence**. Servas has a proven background in software engineering, and is committed to developing innovative AI-driven solutions in healthcare. His work aims to enhance diagnostics, predictive modeling, and personalized medicine through cutting-edge technologies.

SKILLS

Programming Languages: Python, Java, C++
Machine Learning & AI: Tensor Flow, Keras, Scikit-Learn, Yolo, GANs
AI Automation & ChatBot: Lang chain, Natural Language Processing (NLP)
Data Analysis: Healthcare Data Analytics, Statistical Modeling, Multimodal Data Analysis
Tools: Jupyter Notebook, Vscode, GitHub
Healthcare Expertise: HER Data Analysis, White Blood Cell Classification

EDUCATION

Aug. 2023– Present	Soonchunhyang University Ph. D in Big Data Engineering. Advisor: Prof. Woo Ji-Young	Asan, South Korea.
Aug. 2021– Aug. 2023	Soonchunhyang University Ms. in Big Data Engineering. Advisor: Prof. Woo Ji-Young	Asan, South Korea.
Oct. 2017– Aug. 2020	The United African University of Tanzania(UAUT) B.S. in Computer Engineering and Information Technology.	Dar-Es-Salaam, Tanzania
Jan. 2012– Jan. 2014	University of Dar-Es-Salaam Computing Center(UCC). Diploma in Computing and Information Technology.	Dar-Es-Salaam, Tanzania
Jan. 2011– Feb. 2012	University of Dar-Es-Salaam Computing Center(UCC). Certificate in Computing and Information Technology.	Dar-Es-Salaam, Tanzania
Jan. 2007– Nov. 2010	Mbalizi Secondary School Certificate of Secondary School.	Mbeya, Tanzania

RESEARCH EXPERIENCE

Aug. 2021–
Aug. 2023

Soonchunhyang University
Department of Future Convergence
Technology.

Asan, South Korea

Advisor: Prof. Woo Ji-Young

- **White Blood Cell Classification and Segmentation**
Conducted research on the classification and segmentation of white blood cells using machine learning models. This work focused on improving the accuracy of hematological diagnostics through advanced image processing techniques
- **Generative Adversarial Network (GAN's) for Medical Data**
Developed and implemented GANs to generate realistic white blood cell images. This project aimed to augment limited medical datasets with synthetic yet high-quality data, contributing to enhanced machine learning performance in medical diagnostics

Aug. 2023–
Present

Soonchunhyang University
Department of Future Convergence
Technology.

Asan, South Korea

Advisor: Pro. Woo Ji-Young

- **Emotion Detection Using Multimodal Modalities** (Current)
Ongoing project focused on detecting emotions using multimodal data (text, audio, and images) across multiple languages. This work integrates diverse data types to enhance the accuracy of emotion recognition, contributing to the fields of emotional AI and human-computer interaction.
- **Mental Health Analysis**
Developing machine learning models for analyzing mental health conditions. This project focuses on detecting early signs of mental health issues using data-driven approaches, aimed at supporting early intervention and personalized care solutions.

RESEARCH INTEREST

- **Specialized in Multimodal Machine Learning for Healthcare, Medical Data Analysis & Medical Artificial Intelligence.**
- **Multimodal Machine Learning for Healthcare:** I focus on integrating diverse data types (such as text, images, and audio) to enhance healthcare applications. My work involves developing models that can process and analyze multiple modalities simultaneously, improving the accuracy and comprehensiveness of health-related predictions and analyses.
- **Medical Data Analysis:** My expertise centers on extracting actionable insights from complex healthcare datasets, including electronic health records, imaging data, and genomic information. I apply advanced statistical methods and machine learning algorithms to improve diagnostic accuracy, predict patient outcomes, and facilitate personalized medicine. A key project in this area involves white blood cell classification and segmentation, contributing significantly to hematological diagnostics.
- **Generative AI:** I have experience with technologies such as Generative Adversarial Networks (GANs) and other deep learning architectures. My work involves developing and implementing projects that generate new data, images, and text, with a focus on medical applications.
- **Medical Artificial Intelligence:** I specialize in leveraging AI technologies to advance healthcare practices. My experience extends to developing predictive models for disease progression and automating diagnostic processes using sophisticated AI algorithms. A key achievement in this domain is my work on white blood cell classification and segmentation using AI techniques. This project demonstrates my commitment to utilizing AI for enhancing the accuracy and efficiency of medical diagnostics, particularly in hematology.

1. [S.A. Tarimo](#), Mi-Ae Jang, Emmanuel Edward Ngassa, Hee Bong Shing, HyoJin Shin and Jiyoung Woo, "WBC YOLO-ViT: 2 Way – 2 Stage white blood cell detection and classification with a combination of YOLOv5 and vision transformer," *Computer in Biology and Medicine*, Vol. 169, No. 0010-4825, pp. 107875, 2023.
2. Emmanuel Edward Ngassa, Mi-Ae Jang, [S.A. Tarimo](#), Jiyoung Woo and Hee Bong Shing, "Diffusion-based Wasserstein generative adversarial network for blood cell image augmentation," *Engineering Application of Artificial Intelligence*, Vol. 133, No. 0952-1976, pp. 108221, 2024.

INTERNATIONAL CONFERENCE

1. **S.A. Tarimo**, and J. Woo, "White Blood Cell Detection and Classification using YOLOv5 with Hybrid ResNet50-VGG-16-SVM," in Proceedings of the 6th International Conference on ICT for Smart Health & Home (ICT4sHealth & Home), Kota Kinabalu, Malaysia, December 18–22, 2022. (Presentation: December 19).

PATENTS

1. None.

(Open to future innovations and patent opportunities related to my research in AI and healthcare.)

AWARDS AND HONORS

- 2010 Excellence Award – Certificate of Computer Maintenance and Repair, Microsoft Office 2003 and 2007.
- 2010 Excellence Award – Certificate of Network Setup, Repair Internet, Email Usage and Web Design.
- 2012 Excellence Award – Certificate of IT Essentials: PC Hardware and Software (Cisco Networking Academy)
- 2014 Excellence Award – Certificate of IT Essential: PC Hardware and Software (Cisco Networking Academy)
- 2019 Excellence Award – Certificate Software Developer Program and Adobe Photoshop Design.
- 2019 Excellence Award – Certificate of Korea culture Exchange Program (PASS Program).

PARTICIPATING PROJECTS

- | | |
|----------------------|--|
| 2021–2023 | <ul style="list-style-type: none">• White Blood Cell Classification and Segmentation• Generative Adversarial Network (GAN's) for Medical Data |
| 2023– Present | <ul style="list-style-type: none">• Emotion Detection Using Multimodal Modalities• Mental Health Analysis |

REFeree

Prof. [Woo Ji-Young](#), *Ph.D.*
Department of AI and Big Data
Engineering.
Asan, South Korea
jywoo@sch.ac.kr